

Terraform Interview Handbook - 50+ Real Interview Questions (Beginner → Advanced)

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Over the years, I've realized that interviews rarely focus on syntax, they focus on how you think. This handbook is a collection of the Terraform questions I've encountered while preparing for and attending DevOps interviews. I've also included many production and scenario-based questions that interviewers frequently ask. I hope it helps you prepare with more confidence and perform well in your next DevOps interview.

Terraform Basics

1. What is Terraform state and why is it required?
2. Why should we never store terraform.tfstate locally in production?
3. What is the difference between state file and configuration file?
4. Explain terraform init, validate, plan and apply.
5. When would you use terraform refresh?

State Management

6. Your terraform.tfstate file gets corrupted. How do you recover?
7. Two engineers run terraform apply simultaneously. What happens?
8. How does state locking work?
9. Why is Azure Storage Account commonly used as remote backend?
10. How do you migrate local state to remote backend?

Import & Existing Resources

11. Your company already has 200 Azure resources. How will you bring them under Terraform?
12. What happens after running terraform import?
13. Why does terraform import not generate configuration?
14. How do you import an existing AKS cluster?

Modules

15. Why are modules used in production?
16. Root module vs Child module?
17. How do you pass outputs from one module to another?
18. How do you version Terraform modules?
19. How would you structure modules for an enterprise Azure landing zone?

Variables & Locals

- 20. Variable vs Local vs Output?
- 21. Sensitive variable vs normal variable?
- 22. How do you validate that only East US and Central India are allowed regions?
- 23. Difference between nullable and default values?
- 24. How do you generate resource names dynamically?

Loops

- 25. count vs for_each?
- 26. When should you avoid count?
- 27. Scenario: Create 100 VMs with different names.
- 28. Scenario: Create Resource Groups for dev, qa and prod.
- 29. Dynamic block vs for_each?

Lifecycle

- 30. What is prevent_destroy?
- 31. What is ignore_changes?
- 32. Scenario: Azure adds tags manually. Terraform wants to remove them. How do you prevent that?
- 33. create_before_destroy vs replace_triggered_by?
- 34. Scenario: Upgrade Storage Account without downtime.

Dependencies

- 35. How does Terraform determine execution order?
- 36. Explicit dependency vs implicit dependency?
- 37. Which resources run in parallel?
- 38. VM → NIC → Disk → Disk Attachment → Extension. Explain execution order.

Secrets

- 39. How do you securely retrieve secrets from Azure Key Vault?
- 40. Why shouldn't secrets be stored in variables.tf?
- 41. How does OIDC authentication work in GitHub Actions?

42. Service Principal vs Managed Identity vs Federated Identity?

Production Scenarios

43. terraform apply failed after creating 20 resources. What will you do?

44. Terraform wants to recreate a production database after renaming a resource block. How will you avoid downtime?

45. Your teammate manually changed Azure resources. How do you detect drift?

46. Storage Account SKU needs Standard → Premium. Terraform wants recreation. What options do you have?

47. How do you safely deploy infrastructure across two Azure subscriptions?

Azure + Terraform

48. Design Terraform code to deploy:

- Resource Group
- VNet
- Subnet
- AKS
- ACR
- Key Vault

49. Design a GitHub Actions pipeline that:

- Uses OIDC
- Runs Checkov
- Runs terraform fmt
- terraform validate
- terraform plan
- Manual Approval
- terraform apply

50. Design an enterprise Terraform architecture for an AKS platform used by multiple teams. Explain:

- Modules
- Remote Backend
- State Locking
- Environments
- Naming Convention
- CI/CD
- Secrets
- RBAC
- Azure Landing Zone

 Bonus (These are the questions FAANG and Product companies ask)

- Why is Terraform declarative and not imperative?
- Explain Terraform dependency graph.
- What happens internally during terraform apply?
- Why does Terraform sometimes recreate resources?
- How does Terraform detect drift?
- What is a provider plugin?
- Explain terraform graph.
- Explain terraform taint and untaint.
- Explain moved block in Terraform 1.5+.
- What are preconditions and postconditions?
- Explain Checkov integration in CI/CD.
- Atlantis vs GitHub Actions.
- Terragrunt vs Terraform.
- OpenTofu vs Terraform.
- Explain Azure Landing Zone implementation using Terraform.